

Entity & Relationship Analysis Report

E-Commerce Order Management Database System

Document Version: 1.0 | Focus: Data Requirements, Attribute Specification, Constraints & Cardinality Mapping | Target Schema: 3NF Relational Model

1. Executive Summary & System Overview

This report establishes the core conceptual and logical data model for the **E-Commerce Order Management Database System**, fulfilling the Week 2 project deliverable requirements. Building upon the Software Requirements Specification (SRS) defined in Week 1, this analysis bridges business requirements and relational database design.

The primary objective is to identify all necessary entities, define their granular attributes, enforce relational integrity via Primary Keys (PK) and Foreign Keys (FK), and establish domain rules through SQL constraints (NOT NULL, UNIQUE, CHECK, DEFAULT). Furthermore, the report presents a comprehensive relationship and cardinality analysis to ensure seamless data flow across customer management, product cataloging, order processing, inventory tracking, payment verification, and shipment fulfillment.

Key Architectural Highlights

The proposed database schema comprises **9 core entities** designed to operate in strict **Third Normal Form (3NF)**. All many-to-many relationships (e.g., Orders containing multiple Products) have been resolved into associative entities (ORDER_ITEM), ensuring zero data redundancy, high transactional integrity (ACID compliance), and scalable query performance.

2. Entity Identification Report

Based on the functional workflows of an e-commerce platform, nine distinct entities have been identified as essential for capturing the complete lifecycle of an order.

ENTITY NAME	CLASSIFICATION	BUSINESS PURPOSE & JUSTIFICATION
CUSTOMER	Master / Strong Entity	Stores registered user profiles, authentication credentials, and contact details required for order placement and account management.
CATEGORY	Master / Reference Entity	Defines hierarchical product classifications to facilitate catalog browsing, filtering, and inventory reporting.
PRODUCT	Master / Strong Entity	Represents sellable items in the catalog, including pricing, descriptions, SKU identifiers, and category mappings.

ENTITY NAME	CLASSIFICATION	BUSINESS PURPOSE & JUSTIFICATION
INVENTORY	Operational / Weak Entity	Tracks real-time stock levels, warehouse locations, and restock thresholds for each product to prevent over-selling.
ORDER	Transactional Entity	Captures high-level customer purchase transactions, including order timestamps, fulfillment status, and total monetary value.
ORDER_ITEM	Associative Entity	Resolves the M:N relationship between ORDER and PRODUCT. Stores line-item quantities and historical unit prices at the time of purchase.
PAYMENT	Transactional Entity	Records financial transactions associated with orders, tracking payment methods, gateway transaction IDs, timestamps, and payment statuses.
SHIPMENT	Operational Entity	Manages physical delivery logistics, tracking courier details, tracking numbers, dispatch dates, and estimated delivery timelines.
REVIEW	Transactional / Feedback	Captures post-purchase customer feedback, numeric ratings, and moderation status for purchased products.

3. Entity Attribute List & Constraint Specifications

The following tables detail the attributes for each identified entity, specifying their data types, domain constraints, and structural roles within the database.

3.1 Entity: CUSTOMER

Represents individual consumers registered within the e-commerce platform.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
CustomerID	INT / BIGINT	PK NOT NULL AUTO_INCREMENT	Surrogate primary key uniquely identifying each customer.
Email	VARCHAR(150)	UNIQUE NOT NULL	Customer login email and primary communication channel.
PasswordHash	VARCHAR(255)	NOT NULL	Encrypted security hash of the customer's account password.
FirstName	VARCHAR(50)	NOT NULL	Customer's given first name.
LastName	VARCHAR(50)	NOT NULL	Customer's family surname.
Phone	VARCHAR(20)	UNIQUE NULL	Contact telephone number for SMS alerts and delivery verification.
AccountStatus	VARCHAR(20)	NOT NULL CHECK DEFAULT 'Active'	CHECK status IN ('Active', 'Suspended', 'Deactivated').
CreatedAt	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp recording when the account was registered.

3.2 Entity: CATEGORY

Organizes products into logical hierarchy groups for navigation and reporting.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
CategoryID	INT	PK NOT NULL AUTO_INCREMENT	Unique identifier for each product category.
CategoryName	VARCHAR(100)	UNIQUE NOT NULL	Display name of the category (e.g., Electronics, Apparel).
ParentCategoryID	INT	FK NULL	Self-referencing foreign key pointing to `CategoryID` for sub-categories.
Description	TEXT	NULL	Detailed text describing the category scope.
IsActive	BOOLEAN	NOT NULL DEFAULT TRUE	Flag indicating if the category is visible on the storefront.

3.3 Entity: PRODUCT

Stores core information for all sellable goods available on the platform.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
ProductID	INT / BIGINT	PK NOT NULL AUTO_INCREMENT	Unique surrogate identifier for the product catalog item.
SKU	VARCHAR(50)	UNIQUE NOT NULL	Stock Keeping Unit barcode/ alphanumeric code for inventory tracking.
ProductName	VARCHAR(200)	NOT NULL	Title of the product as shown to shoppers.
CategoryID	INT	FK NOT NULL	Foreign key referencing `CATEGORY(CategoryID)`.
UnitPrice	DECIMAL(10, 2)	NOT NULL CHECK	Current selling price. CHECK (UnitPrice >= 0).
WeightGrams	INT	CHECK NULL	Physical weight in grams for shipping cost calculation. CHECK (WeightGrams > 0).
Status	VARCHAR(20)	NOT NULL CHECK DEFAULT 'Active'	CHECK status IN ('Active', 'Draft', 'Discontinued').
CreatedAt	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Date when product was added to the catalog.

3.4 Entity: INVENTORY

Manages real-time stock availability, warehousing, and replenishment triggers.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
InventoryID	INT / BIGINT	PK NOT NULL AUTO_INCREMENT	Unique record identifier for warehouse stock tracking.
ProductID	INT / BIGINT	FK UNIQUE NOT NULL	References `PRODUCT(ProductID)`. Unique enforces 1:1 stock profile per product.
QuantityInStock	INT	NOT NULL CHECK DEFAULT 0	Current physical stock on hand. CHECK (QuantityInStock >= 0).
ReservedStock	INT	NOT NULL CHECK DEFAULT 0	Stock allocated to pending orders not yet shipped. CHECK (ReservedStock >= 0).
ReorderThreshold	INT	NOT NULL DEFAULT 10	Minimum quantity level that triggers automated restock alerts.
WarehouseLocation	VARCHAR(50)	NULL	Aisle/shelf mapping indicator within the fulfillment center.
LastUpdated	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp of the latest stock adjustment or inventory audit.

3.5 Entity: ORDER

Captures customer checkout events and overall transaction fulfillment progression.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
OrderID	BIGINT	PK NOT NULL AUTO_INCREMENT	Unique primary key for the customer order.
CustomerID	INT / BIGINT	FK NOT NULL	References `CUSTOMER(CustomerID)` who placed the order.
OrderDate	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp when the checkout process was completed.
TotalAmount	DECIMAL(12, 2)	NOT NULL CHECK	Final monetary total including taxes and shipping. CHECK (TotalAmount >= 0).
OrderStatus	VARCHAR(30)	NOT NULL CHECK DEFAULT 'Pending'	CHECK status IN ('Pending', 'Processing', 'Shipped', 'Delivered', 'Cancelled').
ShippingAddress	VARCHAR(255)	NOT NULL	Full street address and postal code for physical delivery.
BillingAddress	VARCHAR(255)	NOT NULL	Address associated with the customer's payment method.

3.6 Entity: ORDER_ITEM

Associative entity bridging ORDER and PRODUCT; captures line-item transaction specifics.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
OrderItemID	BIGINT	PK NOT NULL AUTO_INCREMENT	Surrogate primary key for the specific order line item.
OrderID	BIGINT	FK NOT NULL	References `ORDER(OrderID)`. Part of composite unique constraint with ProductID.
ProductID	INT / BIGINT	FK NOT NULL	References `PRODUCT(ProductID)` purchased in this line item.
Quantity	INT	NOT NULL CHECK	Number of units purchased. CHECK (Quantity > 0).
UnitPriceAtPurchase	DECIMAL(10, 2)	NOT NULL CHECK	Historical price per unit at checkout. CHECK (UnitPriceAtPurchase >= 0).
LineTotal	DECIMAL(12, 2)	NOT NULL CHECK	Calculated field: Quantity * UnitPriceAtPurchase. CHECK (LineTotal >= 0).

3.7 Entity: PAYMENT

Records financial settlement details, payment gateway receipts, and transaction status.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
PaymentID	BIGINT	PK NOT NULL AUTO_INCREMENT	Unique identifier for the payment transaction record.
OrderID	BIGINT	FK NOT NULL	References `ORDER(OrderID)`. Connects payment settlement to order.
PaymentMethod	VARCHAR(30)	NOT NULL CHECK	CHECK method IN ('Credit Card', 'Debit Card', 'PayPal', 'UPI', 'Net Banking').
TransactionRef	VARCHAR(100)	UNIQUE NOT NULL	External reference/receipt ID generated by the payment gateway.
AmountPaid	DECIMAL(12, 2)	NOT NULL CHECK	Total funds captured. CHECK (AmountPaid > 0).
PaymentStatus	VARCHAR(20)	NOT NULL CHECK DEFAULT 'Pending'	CHECK status IN ('Pending', 'Completed', 'Failed', 'Refunded').
PaymentDate	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp when the payment gateway processed the transaction.

3.8 Entity: SHIPMENT

Tracks logistics, courier handoff, and physical delivery progression of orders.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
ShipmentID	BIGINT	PK NOT NULL AUTO_INCREMENT	Surrogate primary key for the shipment tracking record.
OrderID	BIGINT	FK NOT NULL	References `ORDER(OrderID)` being fulfilled by this shipment.
CarrierName	VARCHAR(50)	NOT NULL	Logistics provider name (e.g., FedEx, UPS, DHL, BlueDart).
TrackingNumber	VARCHAR(100)	UNIQUE NOT NULL	Courier-assigned waybill/tracking barcode alphanumeric string.
ShipmentStatus	VARCHAR(30)	NOT NULL CHECK DEFAULT 'Packed'	CHECK IN ('Packed', 'Dispatched', 'In Transit', 'Out for Delivery', 'Delivered').
DispatchDate	DATETIME	NULL	Timestamp when package was handed over to carrier.
DeliveryDate	DATETIME	NULL	Timestamp of verified package handoff to the customer.

3.9 Entity: REVIEW

Stores verified buyer product evaluations, numeric ratings, and moderation states.

ATTRIBUTE NAME	DATA TYPE	CONSTRAINTS	DESCRIPTION
ReviewID	BIGINT	PK NOT NULL AUTO_INCREMENT	Surrogate primary key for the product review record.
ProductID	INT / BIGINT	FK NOT NULL	References `PRODUCT(ProductID)` being evaluated.
CustomerID	INT / BIGINT	FK NOT NULL	References `CUSTOMER(CustomerID)` who authored the review.
Rating	SMALLINT	NOT NULL CHECK	Numeric star rating. CHECK (Rating BETWEEN 1 AND 5).
Comment	TEXT	NULL	Optional detailed text feedback provided by the reviewer.
ReviewDate	DATETIME	NOT NULL DEFAULT CURRENT_TIMESTAMP	Timestamp when review was submitted.
IsApproved	BOOLEAN	NOT NULL DEFAULT TRUE	Moderation flag to suppress spam or inappropriate language.

4. Primary Key & Foreign Key Summary Matrix

Relational integrity is enforced across the schema through strict referential links. The following matrix consolidates all primary and foreign key dependencies across the nine entities.

TABLE NAME	PRIMARY KEY (PK)	FOREIGN KEY(S) (FK)	REFERENCED PARENT TABLE(S)
CUSTOMER	CustomerID	None (Master Table)	N/A
CATEGORY	CategoryID	ParentCategoryID	CATEGORY (CategoryID) (Recursive)
PRODUCT	ProductID	CategoryID	CATEGORY (CategoryID)
INVENTORY	InventoryID	ProductID	PRODUCT (ProductID)
ORDER	OrderID	CustomerID	CUSTOMER (CustomerID)
ORDER_ITEM	OrderItemID	OrderID ProductID	ORDER (OrderID) PRODUCT (ProductID)
PAYMENT	PaymentID	OrderID	ORDER (OrderID)

TABLE NAME	PRIMARY KEY (PK)	FOREIGN KEY(S) (FK)	REFERENCED PARENT TABLE(S)
SHIPMENT	ShipmentID	OrderID	ORDER(OrderID)
REVIEW	ReviewID	ProductID CustomerID	PRODUCT(ProductID) CUSTOMER(CustomerID)

5. Relationship & Cardinality Analysis

Understanding the cardinality and participation (optional vs. mandatory) of entity relationships is critical for accurate E-R diagramming and database table generation. Below is the structured analysis of all inter-entity relationships within the order management system.

5.1 Customer places Order

Relationship: Places | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (Order) / Optional (Customer)

Business Rule: A single CUSTOMER can place zero, one, or multiple ORDER records over their customer lifetime. However, every ORDER must be explicitly linked to exactly one valid registered CUSTOMER. Anonymous checkouts are disallowed by schema enforcement (NOT NULL on FK).

5.2 Category contains Product

Relationship: Contains | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (Product) / Optional (Category)

Business Rule: A CATEGORY can encompass multiple sellable PRODUCT items, or remain empty while being set up. Conversely, each PRODUCT must be assigned to exactly one primary classification CATEGORY to ensure organized storefront indexing.

5.3 Order contains Order Items (Associative Resolution)

Relationship: Includes | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (Both Sides)

Business Rule: An ORDER is composed of one or more ORDER_ITEM line items. An order cannot exist without at least one item (enforced via application transaction logic). Each individual ORDER_ITEM is strictly bound to exactly one parent ORDER.

5.4 Product appears in Order Items

Relationship: Listed In | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (OrderItem) / Optional (Product)

Business Rule: A PRODUCT can appear across thousands of distinct ORDER_ITEM records over time as multiple customers purchase it, or zero times if it is a newly listed item. Each ORDER_ITEM record must reference exactly one valid catalog PRODUCT.

5.5 Product tracked in Inventory

Relationship: Has Stock | **Cardinality:** One-to-One (1:1) | **Participation:** Mandatory (Both Sides)

Business Rule: Each sellable PRODUCT corresponds to exactly one active INVENTORY profile that tracks its physical availability, warehouse location, and reservation queues. Enforced by applying a UNIQUE constraint on the foreign key ProductID within the Inventory table.

5.6 Order generates Payment

Relationship: Paid Via | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (Payment) / Optional (Order)

Business Rule: An ORDER typically has one associated PAYMENT. However, a 1:N cardinality is structurally maintained to support real-world payment failure retries, partial payments, split settlements, or post-delivery refund adjustments. Every PAYMENT record must reference an existing ORDER.

5.7 Order fulfilled by Shipment

Relationship: Shipped Via | **Cardinality:** One-to-Many (1:N) | **Participation:** Mandatory (Shipment) / Optional (Order)

Business Rule: While standard orders are fulfilled via a single package (1:1), the system models this as One-to-Many (1:N) to accommodate multi-warehouse split shipments or backordered item dispatches. An ORDER can have zero shipments (if digital or pending), but every SHIPMENT must be tied to exactly one ORDER.

5.8 Customer writes Review for Product

Relationship: Evaluates | **Cardinality:** Many-to-Many (M:N) resolved as (1:N from Customer, 1:N from Product)

Business Rule: A CUSTOMER can author multiple REVIEW records across different products. A PRODUCT can receive multiple reviews from various buyers. The associative table REVIEW bridges this relationship, with unique constraints ensuring a customer can review a specific product only once per verified purchase.

6. Schema Normalization & Verification

The designed schema has been validated against relational database normalization theory to prevent data anomalies (insertion, update, and deletion anomalies):

- **First Normal Form (1NF):** All attributes contain atomic values. Repeating groups (e.g., storing multiple product IDs in a comma-separated string within an order) have been eliminated by creating the individual line-item table ORDER_ITEM.
- **Second Normal Form (2NF):** All tables are in 1NF, and all non-key attributes are fully functionally dependent on the primary key. In ORDER_ITEM (composite identity via OrderID + ProductID), attributes like UnitPriceAtPurchase and Quantity depend entirely on the specific line item, not solely on the Order or the Product.
- **Third Normal Form (3NF):** All tables are in 2NF with zero transitive dependencies. For example, customer contact information is stored exclusively in CUSTOMER and is not redundantly duplicated in ORDER (which only references CustomerID and snapshot shipping addresses).

Deliverable Sign-Off & Verification

This report completes all seven tasks mandated under the Week 2 project scope. The schema is ready for Physical Database Design (SQL DDL execution) in Week 3, where table creation scripts, indexing strategies, and relational triggers will be compiled.